

Non-invasive assessments of the advanced glycation end products in human skin using reflectance NIR spectroscopy

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Abstract — Advanced glycation end products (AGE) are produced by non-enzymatic reaction between glucose and biomolecules including proteins. AGE accumulation is known to cause alternations of structure and function in proteins and to be related with an increased risk of diabetic complications, cardiovascular diseases, and aging processes. Conventionally, AGE accumulation has been estimated by measuring auto fluorescence level using ultraviolet (UV) light excitation. In this study, we investigated an alternative approach to estimate auto fluorescence level and thus AGE accumulation in *in vivo* human skin using NIR (Near-Infrared) spectroscopy. To examine spectral features attributed to glycation in proteins, we first analyzed *in vitro* NIR spectra from native and glycated protein. Then, we further examined NIR spectra of *in vivo* skin from human subjects, and estimated their auto fluorescence level using several multivariate regression approaches. Our analysis in *in vitro* spectra from native and glycated albumin revealed that glycation may affect –CH and –NH stretching. Furthermore, we elucidated that those bands for –CH and –NH may be responsible for the variation in auto fluorescence level in human skin NIR spectra. Finally, auto fluorescence level was estimated from those NIR spectra using several multivariate regression methods: principal component regression (PCR), partial least square regression (PLS-R) and support vector regression (SVR). Among the three methods, SVR showed the best performance. We demonstrated in this study that NIR spectroscopy can be used as an alternative non-invasive method to estimate AGE accumulation in *in vivo* human skin tissue without UV radiation on skin tissue.

I. INTRODUCTION

Advanced glycation end products (AGE) are produced by non-enzymatic reaction. Glucose is attached to protein, lipids or nucleic acid through Maillard reaction. These structural modifications affect molecular conformations of bio molecules by increasing cross-linking. The formation of cross-links contributes to the alterations of functional and physical features of proteins [1]. AGE accumulation in tissues is known to be related to the progression of aging, cardiovascular disease and diabetic complications [2]. The amount of AGE in skin tissue is important to be measured to assess the risk of diabetes and diabetic complications. Diabetic patients exhibit the increased accumulation of AGE when compared with normal subjects. Aging, diabetes and diabetic complications enhance the accumulation of AGE in skin tissue [3]. To assess the accumulation of AGE in human skin tissue, auto fluorescence (AF) has been measured. Ultraviolet light excites AGE molecules in skin tissue and the emitted fluorescence from skin is measured. AGE analyzer based on skin auto

fluorescence has been developed and used to assess the risk for cardiovascular disease, diabetes and diabetic complications [4].

Alternatively, vibrational spectroscopy including near-infrared (NIR) spectroscopy and Raman spectroscopy has been used to analyze protein glycation in human. Fingernails from non-diabetic and diabetic patients were analyzed using NIR spectroscopy. NIR spectroscopy combined with partial least square discriminant analysis (PLS-DA) showed its potential for a non-invasive method to diagnose diabetic mellitus [5]. Peak ratio in Raman spectra from diabetic and non-diabetic patients exhibited the correlation with skin fluorescence. Raman spectroscopy was reported for its capability to screen diabetes [6].

The aim of this study was to provide a non-invasive method to assess auto fluorescence level based on the deposition of AGEs in skin and to verify whether NIR spectroscopy can be an alternative method for the assessment of the risk for aging, diabetic complications and cardiovascular disease. First, NIR spectra from native and glycated protein were examined to elucidate spectral features attributed to glycation in proteins. Then, we analyzed NIR spectra from human subjects to assess auto fluorescence level and estimated the accumulation of AGEs in skin tissue. To estimate auto fluorescence level resulted from the accumulation of AGEs in skin, prediction models were established based on multivariate regression method including principal component regression (PCR), partial least square regression (PLS-R) and support vector regression (SVR).

II. MATERIALS AND METHODS

Albumin and glycated albumin were purchased from Sigma-Aldrich (St. Louis, U.S.A.). Albumin and glycated albumin were mixed with potassium bromide powder to make pellets. NIR transmittance spectra were obtained using VERTEX 80 FTIR spectrometer (Bruker, Germany). AGE accumulation in skin was measured based on auto fluorescence using AGE reader (DiagnOptics, Groningen, Netherlands) as reference. NIR spectra from human skin were obtained using VERTEX 80 FTIR spectrometer in spectral region from 7000 cm^{-1} to 4000 cm^{-1} at 16 cm^{-1} resolution with an InGaAs detector operating throughout entire region. A light source was a halogen lamp. The power of light was 150 watt. One spectrum was obtained by averaging after 64 scanning. A fiber optic probe (Bruker, Germany) was used to acquire reflectance spectrum on human skin. This optic probe contains a bundle of optical fibers with illumination fibers and detecting fibers.

Fourteen volunteers (ten males and four females, mean age of 37.3-year old) without any dermatological disorder participated in our study. The volunteers were asked not to use any skin care products or medications on their volar forearms for 24 hours prior to study.

A reference spectrum for absorbance calculation was obtained using Spectralon (Labsphere, New Hampshire). Multiplicative Scatter Correction (MSC) method was applied for spectrum pre-processing. Savitzky-Golay 2nd derivative was performed before the multivariate regression analysis of spectra obtained from skin. The second derivatives of skin spectra were calculated with nine points fitted to third-order polynomial to examine the structural feature of proteins in the skin tissue. After preprocessing spectra, regression models to predict auto fluorescence level in skin were established using PCR, PLS and SVR methods. The R language (version 3.5.1) and Panorama (version 3.3.4, Lab Cognition, Germany) were used for spectral analysis and statistical computing. Packages in R language including 'pls', 'e1071' and 'prospectr' were installed for the spectral analysis. The experimental procedure involving human subjects in this study followed the principles in the Helsinki Declaration of 1975, as amended in 2000.

III. RESULTS AND DISCUSSION

NIR spectra of native and glycated albumin were examined to explain how glycation affects spectral features. Transmittance NIR spectra of native and glycated albumin in the region between 4000cm⁻¹ and 7000 cm⁻¹ are shown in Figure 1. Spectral variations were observed in peaks at 4500 cm⁻¹, 5100 cm⁻¹, 5740 cm⁻¹ and 6610 cm⁻¹. Especially, the band at near 5740 cm⁻¹ is assigned for the first overtone of the -C-H stretching mode, and the band at 6610 cm⁻¹ is assigned for the first overtone of the -N-H stretching. The band at 5100 cm⁻¹ is assigned for the combination of -N-H stretching and bending [7]. These results suggest that glycation in protein may affect secondary structure of protein leading to changes in -C-H and -N-H stretching.

To elucidate bands contributing to spectral variations in skin NIR spectra, principal component analysis (PCA) was conducted. Principal component (PC) one and two explained 95.67 % of spectral variations. Figure 2 shows PC1 and PC2 loadings in spectral region of 7000 cm⁻¹ to 4000 cm⁻¹. In PC2

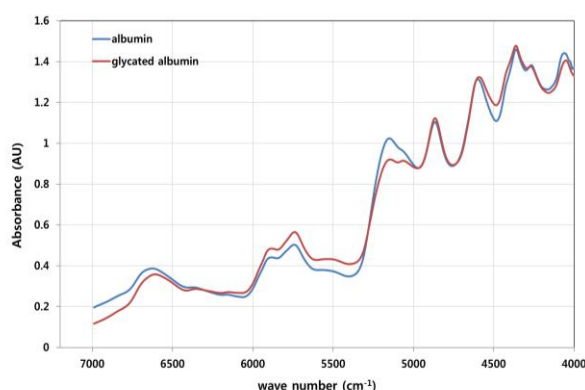


Figure 1. NIR spectra of albumin (blue) and glycated albumin (red).

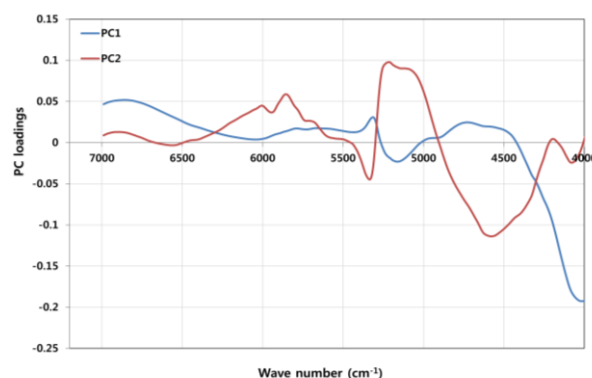


Figure 2. PC1 (blue) and PC2 (red) loadings of NIR reflectance skin spectra obtained from each subject.

loadings, peaks were shown at near 5100 cm⁻¹ and 5700 cm⁻¹. As shown in Figure 1, the peak at 5100 cm⁻¹ is assigned for -N-H stretching and bending, and the peak at 5700 cm⁻¹ is assigned for -C-H stretching. These peaks contributed to variation in spectra caused by structural changes in protein in skin tissue. This result suggests that changes in secondary structure of protein in skin tissue mainly contribute to individual spectral variation. Considering spectral variation caused by glycation in protein, spectral variations in human skin tissue can be explained by the variations in the levels of glycation in skin stratum corneum, epidermis and dermis.

Savitzky-Golay second derivatives of spectrum were calculated to investigate structural changes in proteins by reducing baselines [5]. The second derivatives of NIR spectra obtained from human and mouse skin were analyzed to assess the chronological change in skin [8, 9]. In this study, we performed second derivative analysis of spectra obtained from human subjects. Derivative values at each wave number in the second derivative spectra obtained from each subject were investigated to calculate the correlation coefficients with auto fluorescence level of each human subject. Auto fluorescence level was measured using AGE reader. The Pearson correlation coefficients at each of the wave number between second derivative value of spectra and skin auto fluorescence level obtained from each subject are shown in Figure 3. The maximum coefficient was shown at 7000 cm⁻¹. However, the spectra at this wave number contain more noise levels. The region between 4200 cm⁻¹ and 5100 cm⁻¹ showed the

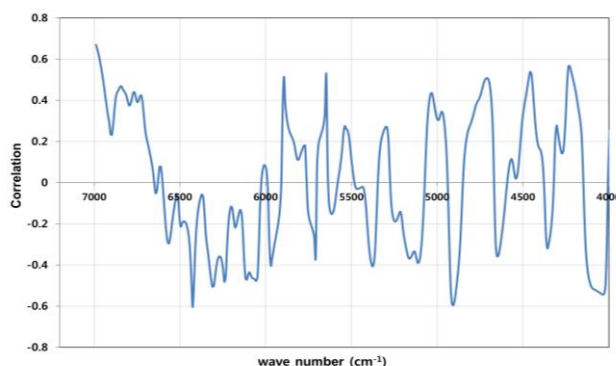


Figure 3. The Pearson correlation coefficients at each wave number between second derivative of spectra and auto fluorescence level obtained from each subject.

correlations over 0.5. Bands in this region are assigned to combination band for -C-H vibration and -N-H stretching in protein [9]. In addition, there were two other bands showing correlation coefficients over 0.5 in the region between 5500 cm^{-1} and 6000 cm^{-1} . Bands in this region are related with the first overtone of -C-H groups. In the second derivative spectra obtained from human skin, changes in these bands are related with physiological aging in human skin [9]. These results imply that the spectral changes in the region of 4200 cm^{-1} ~ 5100 cm^{-1} and 5500 cm^{-1} ~ 6000 cm^{-1} is related with physiological aging accompanied by the accumulation of AGE in skin tissue.

In order to build calibration models to predict the accumulation of AGE in human skin based on the reflectance NIR spectroscopy, multivariate regression including PCR, PLS-R and SVR were performed. Before the establishment of calibration models, NIR spectra were pre-processed with MSC and Savitzky-Golay 2nd derivative method. Validation of calibration model and prediction of AF based on PCR method were shown in Figure 4. PCR method is to use the principal components scores rather than the original data for the regression. Left panel in Figure 4 shows the RMSEP (root mean square error of prediction) estimate against the number of PCs in the calibration model. Leave-one-out cross validation was performed to assess the optimal number of PCs. Nine PCs exhibited the best performance in prediction. Therefore, nine PCs were used to calculate prediction model. Right panel in Figure 4 shows the prediction of auto fluorescence for fourteen subjects. The correlation coefficients between measured and predicted AF was 0.695 and RMSEP in prediction was 0.20 (a.u. arbitrary unit for AF).

PLS regression uses orthogonal latent variables to reduce dimension of original data for the regression. Validation of calibration model and prediction of AF based on PLS-R method are shown in Figure 5. Left panel shows the RMSEP estimate vs the number of latent variables (LV) in calibration model. Leave-one-out cross validation was performed to obtain the optimal number of LVs. Calibration model using 7 LVs exhibited the best performance in the prediction of AF. Right panel shows the prediction AF based on PLS-R calibration model. The correlation coefficient between measured and predicted AF value was 0.733 and RMSEP in prediction was 0.183 (a.u.). The calibration model based on

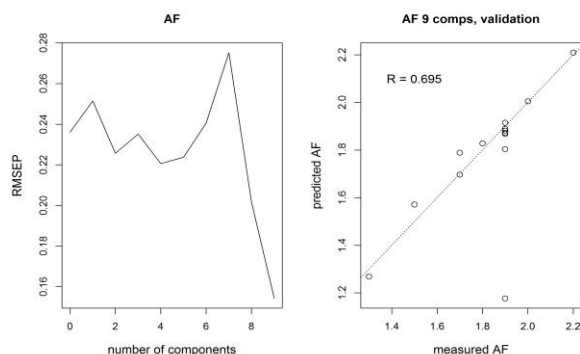


Figure 4. Validation plot for PCR on the NIR spectral data obtained from human skin (left) and prediction quality of the optimal model for fourteen subjects, containing nine PCs (right). AF denotes auto fluorescence level in skin.

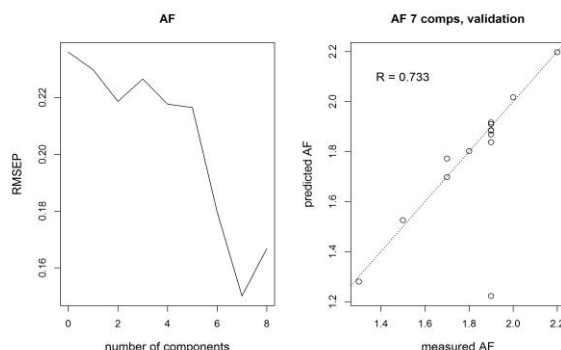


Figure 5. Validation plot for PLS-R on the NIR spectral data obtained from human skin (left) and prediction quality of the optimal model for fourteen subjects, containing seven latent variables (LV) (right). AF denotes auto fluorescence level in skin.

PLS-R exhibited the better result in the prediction of AF based on skin NIR spectra with the lower number of LVs when it was compared with PCR.

Support vector machine (SVM) is a classification method and extended for regression analysis in chemometrics. SVR has shown the advantage in dealing with ill-posed problems and the better performance than regression method like PLS in developing calibration models [10]. To establish the best calibration model using SVR, optimization process is required by tuning parameters for allowable error (epsilon) and cost. The result of optimization for calibration model based on SVR is shown in Figure 6. The epsilon was 0.2 and the cost was 4.0 to minimize mean square error (MSE).

The SVR calibration model was established using parameters obtained by the optimization process. Figure 7 shows the prediction result of AF based on SVR method. The correlation coefficient was 0.971 and RMSEP in prediction was 0.061 (a.u.). The calibration model established by SVR exhibited the improved prediction result compared with linear regression method such as PCR and PLS-R. The SVR calibration model relies on non-linear kernel function (radial basis function) to construct the model. The kernel function transforms the training data from non-linear space to linear space. This non-linear kernel function may lead to improve the prediction result. Skin is a complex system containing macromolecules such as protein and lipid. This complexity can cause the non-linearity in spectra. The reduction of non-linearity by kernel function may improve the performance of SVR in predicting AF based on spectra.

In this study, it was demonstrated that glycation affected -C-H and -N-H bands in NIR spectra and these bands are related with inter-subject variation in skin auto fluorescence. The modification of -C-H and -N-H by glycation may contribute to the structural changes in proteins leading to the elevation of auto fluorescence in skin tissue.

IV. CONCLUSION AND FUTURE WORKS

In this study, we demonstrated an alternative approach to estimate AGE accumulation on skin tissue by using NIR, which was done without UV radiation on skin. The three regression methods were examined to assess AGE

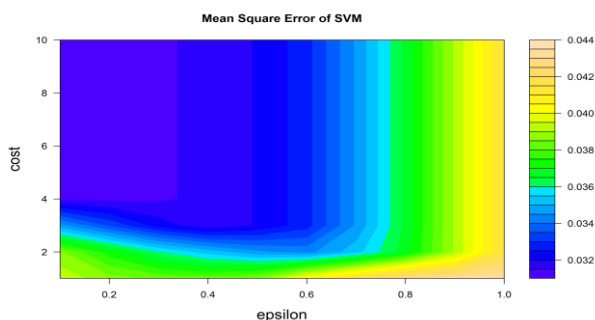


Figure 6. Optimization of SVR based on the NIR spectral data obtained from human skin. The optimal parameters, epsilon and cost, were chosen to minimize mean square error.

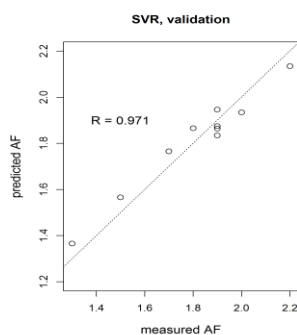


Figure 7. The prediction of AF using SVR model for fourteen subjects.

accumulation in human skin based on reflectance NIR spectroscopy. SVR exhibited the best performance in the prediction of auto fluorescence level in skin tissue.

The glycation of proteins results in the formation of cross-link in the extracellular matrix leading to a decrease in the elasticity and an increase in vessel stiffness. These physical changes in extracellular matrix proteins result in the development of cardiovascular disease and diabetic complications. AGE also affects vascular cells including endothelial cells and pericytes through the activation of receptor for AGE in the pathogenesis of diabetic retinopathy [11]. Furthermore, it has been known that glycation-induced structural changes in proteins are related with aging in skin, vascular aging, immune dysfunction and cerebral aging [12]. Hence, the method to monitor auto fluorescence level based on the contents of AGE can provide to assess aging and the onset of diabetic complications. This study suggests that NIR reflectance spectroscopy can be a non-invasive method to evaluate aging level, the onset of diabetic complications and the risk of cardiovascular disease by the prediction of auto fluorescence level in skin without UV radiation on skin.

This study was conducted on the limited number of normal subjects without any disease. The assessment of auto fluorescence level on a large number of patients with metabolic disease and normal subjects without metabolic disease using reflectance NIR spectroscopy is required for the further study. This further study may provide a non-invasive diagnostic tool for diabetes, diabetic complications and aging based on changes in protein structure resulted from the accumulation of AGE in cellular matrix. This diagnostic tool may evaluate the efficacy of medication for the disorder in

metabolism and cardiovascular system by evaluating changes in protein glycation.

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